

条件极值

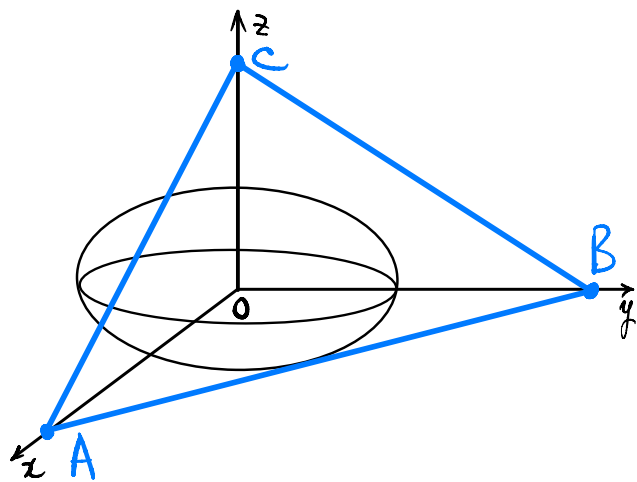
例. 作椭球面

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

($a, b, c > 0$) 之切平面在

第一卦限之切平面与坐

标平面围成一个四面体. 求该四面体的最大体积



解. 设 $P_0(x_0, y_0, z_0)$ 在椭球面上, 则 (x_0, y_0, z_0) 满足约束条件

$$g(x_0, y_0, z_0) \triangleq \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2} - 1 = 0$$

$$g_x(P_0) = \frac{2x_0}{a^2}, \quad g_y(P_0) = \frac{2y_0}{b^2}, \quad g_z(P_0) = \frac{2z_0}{c^2}.$$

椭球面 $g(x, y, z) = 0$ 在 P_0 处的法向量为

$$\vec{n} = \{g_x(P_0), g_y(P_0), g_z(P_0)\} = \left\{ \frac{2x_0}{a^2}, \frac{2y_0}{b^2}, \frac{2z_0}{c^2} \right\}.$$

所以在 P_0 处的切平面方程为

$$\text{切平面方程: } \frac{2x_0}{a^2}(x-x_0) + \frac{2y_0}{b^2}(y-y_0) + \frac{2z_0}{c^2}(z-z_0) = 0$$

化简为

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} + \frac{z_0 z}{c^2} = 1. \quad \text{—— 切平面方程.}$$

切平面在 x, y, z 上的截距为:

$$x = \frac{a^2}{x_0} = OA \quad y = \frac{b^2}{y_0} = OB, \quad z = \frac{c^2}{z_0} = OC$$

四面体的体积 V :

$$V = \frac{1}{6} ABC = \frac{1}{6} \frac{a^2 b^2 c^2}{x_0 y_0 z_0}$$

求 V 的最小值就是求

$V(x_0, y_0, z_0)$ 在约束条件

$$g(x_0, y_0, z_0) \triangleq \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2} - 1 = 0$$

下的条件极值.

常规方法: 作 Lagrange 函数

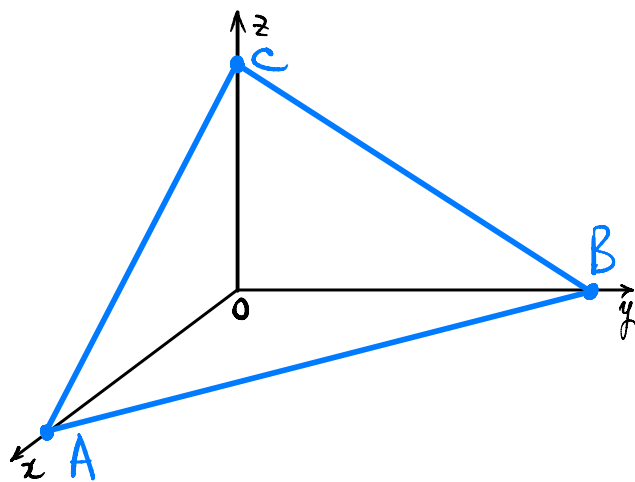
$$L = V - \lambda g = \frac{a^2 b^2 c^2}{x_0 y_0 z_0} - \lambda \left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2} - 1 \right).$$

但偏导数 $\frac{\partial L}{\partial x_0}, \frac{\partial L}{\partial y_0}, \frac{\partial L}{\partial z_0}$ 相对复杂, 还有更简单的方式.

事实上, 求 $V = \frac{1}{6} \frac{a^2 b^2 c^2}{x_0 y_0 z_0}$ 在 $g(x_0, y_0, z_0) = 0$ 下的极值,

等价于求 $\ln V$ 在 $g(x_0, y_0, z_0) = 0$ 下的极值

$$L(x_0, y_0, z_0, \lambda) = \ln x_0 + \ln y_0 + \ln z_0 - \lambda \left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2} - 1 \right).$$



$$\begin{aligned}
 \frac{\partial L}{\partial x_0} &= \frac{1}{x_0} - \lambda \frac{2x_0}{a^2} = 0 \\
 \frac{\partial L}{\partial y_0} &= \frac{1}{y_0} - \lambda \frac{2y_0}{b^2} = 0 \\
 \frac{\partial L}{\partial z_0} &= \frac{1}{z_0} - \lambda \frac{2z_0}{c^2} = 0 \\
 g &= 0
 \end{aligned}
 \quad \leftarrow \quad
 \begin{cases}
 1 - \lambda \frac{2x_0^2}{a^2} = 0 \\
 1 - \lambda \frac{2y_0^2}{b^2} = 0 \\
 1 - \lambda \frac{2z_0^2}{c^2} = 0 \\
 \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2} = 1
 \end{cases}$$

$$\Rightarrow \frac{x_0^2}{a^2} = \frac{y_0^2}{b^2} = \frac{z_0^2}{c^2} = \frac{1}{2\lambda}$$

$$\Rightarrow \frac{x_0^2}{a^2} = \frac{y_0^2}{b^2} = \frac{z_0^2}{c^2} = \frac{1}{3}$$

$$\Rightarrow x_0 = \frac{a}{\sqrt{3}}, \quad y_0 = \frac{b}{\sqrt{3}}, \quad z_0 = \frac{c}{\sqrt{3}}$$

$\Rightarrow$ 四面体最小体积为

$$V = \frac{1}{6} \frac{a^2 b^2 c^2}{x_0 y_0 z_0} = \frac{a^2 b^2 c^2}{6} \frac{1}{\frac{a}{\sqrt{3}} \frac{b}{\sqrt{3}} \frac{c}{\sqrt{3}}} = \frac{\sqrt{3}}{2} abc$$

$\therefore$ 最小体积

$$\min V(x_0, y_0, z_0) = \frac{\sqrt{3}}{2} abc, \quad \#$$

$$F(x_0, y_0, z_0) = 0$$

第十六章 隐函数存在定理

给定 x , 从 $F(x, y) = 0$, 求 $y = y(x)$. —— 隐函数
以前已学.

“如果” $y = y(x)$ 大前提 $\exists$ 且可导, 且 $F_y \neq 0$. “那么”:

$$F_x + F_y \cdot y'(x) = 0 \Rightarrow y'(x) = -\frac{F_x}{F_y}$$

先决条件是 $F_y \neq 0$: $\begin{cases} \text{要么 } F_y > 0, & F(x, y) \text{ 关于 } y \nearrow; \\ \text{要么 } F_y < 0, & F(x, y) \text{ 关于 } y \searrow; \end{cases}$

方程组情况:

$$\begin{cases} g(x, y, u, v) = 0 \\ h(x, y, u, v) = 0 \end{cases}$$

“如果” 大前提 $\exists$ 隐函数组 $\begin{cases} u = u(x, y), \\ v = v(x, y), \end{cases}$

且 $\frac{\partial(g, h)}{\partial(u, v)} \neq 0 \Rightarrow$ 可求出 u, v 的偏导数.

那么, 什么条件下, 隐函数(组)存在呢?

方法一: 用单调性; 方法二: 迭代格式.

例如 ODE 解的存在唯一性定理之证明:

考虑 ODE:
$$\begin{cases} \frac{dy}{dx} = f(x, y), & (x, y) \in D, \\ y(x_0) = y_0 \end{cases}$$

D 是矩形: $|x-x_0| \leq a, |y-y_0| \leq b$.

条件: $f(x,y) \in C(D)$. 关于 y 是 Lip 连续.

$$\text{则 } ODE \Leftrightarrow y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt.$$

$$\text{作映射: } v(x) \mapsto u(x) = Tv(x) \triangleq y_0 + \int_{x_0}^x f(t, v(t)) dt$$

$$y = \varphi(x) \text{ 是解} \Leftrightarrow \varphi(x) \text{ 是映射 } T \text{ 之不动点: 即: } \\ T\varphi(x) = \varphi(x).$$

由方程 $F(x, y) = 0, (x, y) \in D$, 求隐函数.

$$? \quad y(x), \quad \exists \quad F(x, y(x)) \equiv 0?$$

$$\text{等价地: } \forall A \neq 0 \quad y(x) - AF(x, y(x)) \equiv y(x).$$

$$\text{作映射 } T: v(x) \mapsto u(x) = Tv(x) \triangleq v(x) - AF(x, v(x))$$

$$y(x) \text{ 是隐函数} \Leftrightarrow y \text{ 是 } T \text{ 之不动点: } Ty(x) = y(x).$$

构造 Picard 迭代格式: 取 $y_0(x)$. 令

$$y_1(x) = Ty_0(x) = y_0(x) - AF(x, y_0(x)).$$

$$y_{n+1}(x) = Ty_n(x) = y_n(x) - \underbrace{AF(x, y_n(x))}_{\downarrow}$$

$$\downarrow \leftarrow \text{证明之} \quad \downarrow \quad \downarrow \\ y_*(x) \quad \quad \quad = y_*(x) - AF(x, y_*(x))$$

$\Rightarrow y_*(x)$ 就是隐函数

定理1.1 (隐函数存在定理). 设 $D = [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b]$,

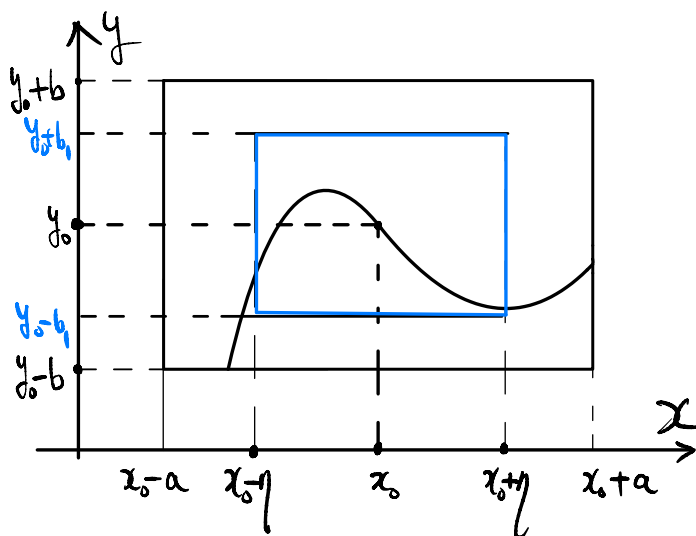
$F(x, y)$ 是 D 上的函数, 满足

(1) $F(x, y) \in C^1(D)$. 即

F_x, F_y 在 D 中连续.

(2) 曲线 $F(x, y) = 0$ 过 (x_0, y_0) ,

即 $F(x_0, y_0) = 0$,



(3) $F_y(x_0, y_0) \neq 0$.

则 $\exists \eta \in (0, a]$: $\Rightarrow F(x, y) = 0$ 确定了唯一的隐函数

$y = f(x)$, $|x - x_0| \leq \eta$. 满足下列性质:

(i) $y = f(x) \in C^1[x_0 - \eta, x_0 + \eta]$. 且

$y_0 = f(x_0)$; $(x, f(x)) \in D$; $\forall x \in [x_0 - \eta, x_0 + \eta]$.

(ii) $F(x, f(x)) = 0$. $\forall x \in [x_0 - \eta, x_0 + \eta]$.

(iii) $\frac{df(x)}{dx} = -\frac{F_x}{F_y}$. $|x - x_0| \leq \eta$

证明: Step 1: 设曲线方程为 $y = f(x)$.

则曲线方程 $y = f(x)$: $\Rightarrow F(x, f(x)) \equiv 0$, $|x - x_0| \leq \eta$:

等价于: 取适当的 $A \neq 0$: $\Rightarrow$

$$f(x) \equiv f(x) - A F(x, f(x)), |x - x_0| \leq \eta.$$

定义一个 "Picard 迭代格式": 令 $y_0(x) = y_0$.

$$y_1(x) = y_0(x) - A F(x, y_0(x)).$$

$(x_0, y_0) \in D$, $\checkmark$

$$y_2(x) = y_1(x) - A F(x, y_1(x)).$$

$\vdots$

$$y_n(x) = y_{n-1}(x) - A F(x, y_{n-1}(x)) : \quad \forall n \geq 1.$$

还要保证, $(x, y_{n-1}(x)) \in D$.

如要证明: $y_n(x) \rightarrow f(x)$. 则

$$\begin{array}{ccc} y_n(x) & = & y_{n-1}(x) - A F(x, \underbrace{y_{n-1}(x)}_{f(x)}) \quad (\because F(x, y) \text{ 连续}) \\ \downarrow & & \downarrow \quad \downarrow \\ f(x) & = & f(x) - A F(x, f(x)). \end{array}$$

即: $y = f(x)$ 确实是隐函数.

再证明 $f(x)$ 可导. 于是得到 $f'(x) = -\frac{F_x}{F_y}$.

Step 2: 估计 $y_n(x)$: $\exists k \in (0, 1)$:

$$|y_{n+1}(x) - y_n(x)| \leq k |y_n(x) - y_{n-1}(x)|, \quad |x - x_0| \leq \eta.$$

$$y_{n+1}(x) = y_n(x) - A F(x, y_n(x))$$

$$\rightarrow y_n(x) = y_{n-1}(x) - A F(x, y_{n-1}(x))$$

相减得:

$$\begin{aligned} y_{n+1}(x) - y_n(x) &= (y_n(x) - y_{n-1}(x)) - A [F(x, \underline{y_n(x)}) - F(x, y_{n-1}(x))] \\ &= \underline{(y_n(x) - y_{n-1}(x))} - A F_y(x, \xi_n(x)) (y_n(x) - y_{n-1}(x)) \\ &= (y_n(x) - y_{n-1}(x)) [1 - A F_y(x, \xi_n(x))]. \end{aligned}$$

这是因为 $F(x, y)$ 关于 y 可导, 由 Lagrange 中值定理,

$\exists \xi_n(x)$: 介于 $y_{n-1}(x)$ 与 $y_n(x)$ 之间,

$$\therefore |y_{n+1}(x) - y_n(x)| = |1 - A F_y(x, \xi_n(x))| \cdot |y_n(x) - y_{n-1}(x)|.$$

$$\because F_y(x_0, y_0) \neq 0, \text{ 取 } A \neq 0: \exists A \cdot F_y(x_0, y_0) = 1$$

而 $1 - A F_y(x, y)$ 关于 x, y 连续.

$$\therefore \exists (x_0, y_0) \text{ 邻域 } D_1 = \{(x, y) \mid |x - x_0| \leq \eta_1, |y - y_0| \leq b_1\}$$

(即 $\exists \eta_1 \in (0, a], b_1 \in (0, b]$) 使得

$$|1 - A F_y(x, y)| < \frac{1}{2} \triangleq k, \quad \forall (x, y) \in D_1.$$

如果在上述迭代过程中, 始终保证有

$$(x, y_n(x)) \in D_1, \text{ 则}$$

← 下面证明之

$$|1 - AF_y(x, \xi_n(x))| \leq k = \frac{1}{2},$$

$\Rightarrow \forall n \geq 1:$

$$|y_{n+1}(x) - y_n(x)| \leq \frac{1}{2} |y_n(x) - y_{n-1}(x)|, \quad \forall |x - x_0| \leq \eta_1.$$

类似于 ODE 解的存在唯一性定理的证明过程，
可知 $\{y_n(x)\}$ 是 Cauchy 列（连续函数的基本列）。

$$|y_n(x) - y_m(x)| \xrightarrow{m, n \rightarrow \infty} 0$$

从而 $\{y_n(x)\}$ 是一致收敛的：

其极限就是欲求之连续函数。

作业：P159. 2. 4.